

Latent-Space Diagnostics: Push-T

The same diagnostics as Two-Room, on Push-T’s encoder and predictor geometry – and where the two environments genuinely differ

Companion to `predictor-stitching.tex` and `latent-diagnostics.tex`

6 September 2026

1 Purpose

The same diagnostics as `latent-diagnostics.tex`, re-run on Push-T’s 300-episode B0 landmark set (`scripts/latent_diagnostics.py env=pusht`), reusing the existing cached landmarks (`outputs/b0_pusht/landmarks_ep300_seed0.npz`) and the real `quentinll/lewm-pusht` checkpoint. Several of these numbers land close to Two-Room’s; the geometry (adjacent-pair scale, cosine similarity) genuinely differs, and lines up directly with `main.tex`’s own finding that Push-T’s calibration formula fails outright where Two-Room’s does not. The predictor-accuracy figures (Figure 3–Figure 6) were originally measured incorrectly twice over – first the wrong action-block shape, then unnormalized actions – and have since been corrected both times; see the two correction notices before Figure 3.

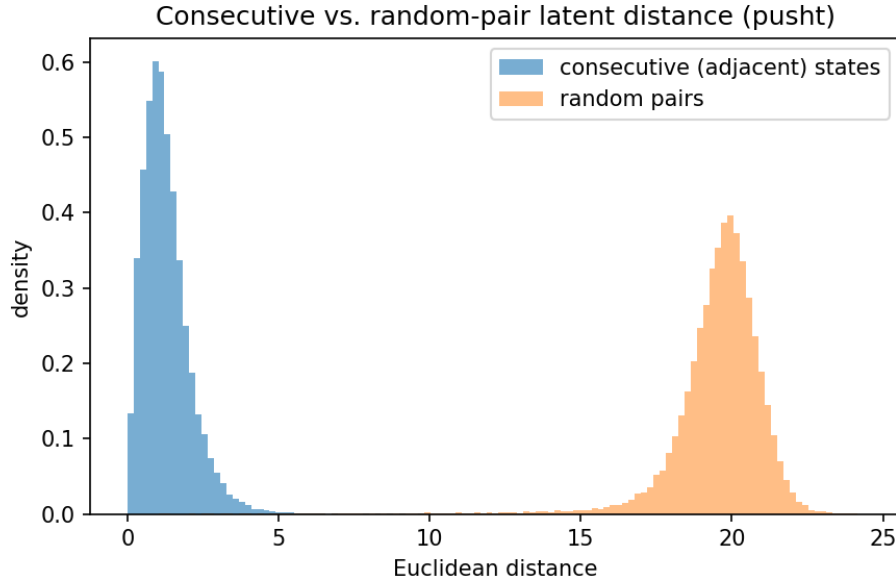


Figure 1: Consecutive states sit at mean 1.28 (median 1.15), a full order of magnitude below Two-Room’s 11.75. Random pairs are comparable to Two-Room’s (19.57 vs. 19.25) — so the gap between “one real step” and “two arbitrary states” is far larger in Push-T’s embedding than in Two-Room’s. This alone is a strong hint that Push-T’s dynamics move the encoder’s output much more slowly per step, consistent with slow, continuous contact manipulation rather than Two-Room’s larger free-space steps.

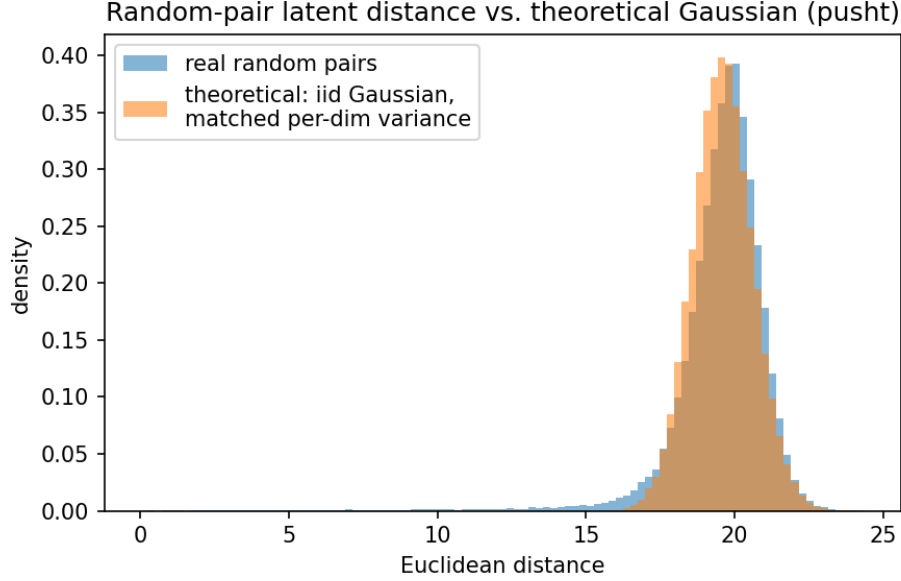


Figure 2: Real random-pair distances (mean 19.57) closely track the theoretical matched-variance Gaussian reference (mean 19.60) — as close a match as Two-Room’s (19.25 vs. 19.32). At the level of pairwise distance statistics alone, Push-T’s embedding looks just as Gaussian-like in aggregate as Two-Room’s; whatever breaks `main.tex`’s calibration formula on Push-T, it is not visible in this particular statistic.

Correction (6 September 2026): the four items below replace an earlier, wrong analysis.

The original version of this section fed the predictor a single real action, zero-padded into the 10-dim slot `action_encoder` expects, and compared its output one real step later – and found what looked like a predictor performing at roughly chance level. That invocation was wrong, not the model. The checkpoint’s actual training convention (`docs/lewm.txt`, Appendix D) applies a **frame-skip of 5**: one predictor call consumes the concatenation of **5 consecutive real actions** ($5 \times 2=10$ dims, exactly matching `action_encoder`’s input size with no padding at all) and predicts the state **5 real steps** later, not 1. Three specific bugs were ruled out before this deeper one was found – wrong action-encoder slot (all 10 zero-padding positions land in the same narrow MSE band), the action pathway being ignored (predictions do vary with different real actions), and off-by-one frame/action alignment (`action[i]` genuinely beats `action[j]`) – see `[[lewm-action-encoder-padding]]` in memory for that record. None of those three was the actual explanation; the block-size mismatch was.

Second correction (9 September 2026): the 6 September numbers above were themselves still wrong. The block-size fix was correct, but the actions feeding `action_encoder` were still *raw*, not z-scored. The checkpoint consumes z-scored actions on both sides of the pipeline (`train.py`’s column normalizer, `eval.py`’s `StandardScaler`) – see `CLAUDE.md`, “Action normalization,” and the companion correction in `latent-horizon-pusht.tex`. Push-T’s action std is ≈ 0.206 , so raw actions are $4.82\times$ too small, which reads to the predictor as a near-stationary command – exactly the “flat, no history benefit, but otherwise unremarkable” picture the 6 September numbers showed. Figures 3–6 below are regenerated with z-scored actions (`scripts/latent_diagnostics.py env=pusht`, action stats computed over the full dataset action column, same convention as `idm_gate.py`); the 6 September values they replace are kept in the

running text below for comparison, labeled as such.

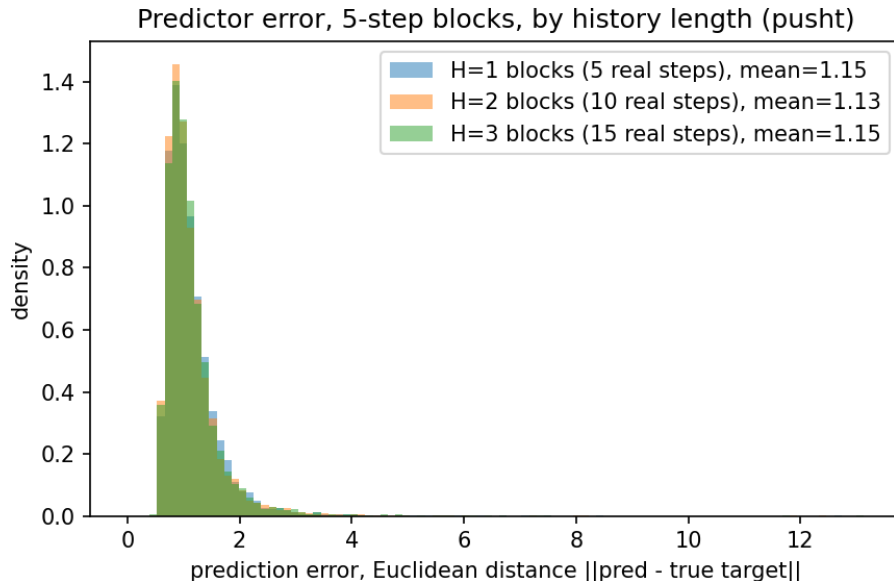


Figure 3: Predictor error with z-scored actions (5-action-block convention, corrected 9 September), at $H=1, 2, 3$ blocks of history (5, 10, 15 real steps of context), on 5000 real chains per length: mean 1.152, 1.133, 1.149 (median 1.025, 0.996, 1.012; q90 1.675, 1.631, 1.662). Error is still flat across H – still no benefit from Push-T’s documented history length of 3 on this metric, the one genuine surprise that survives both corrections. But the 6 September (raw-action) draft reported means 4.27, 4.36, 4.42 – roughly $3.7\times$ larger, and close to the raw 5-step real-movement scale itself (Figure 1 scaled up), i.e. a near no-motion error. With actions correctly normalized, mean error 1.152 sits right at the real single-step movement scale (1.15), not at the movement scale of the whole 5-step block – the predictor is now explaining most of a block’s real displacement, not almost none of it.

predictor error relative to real movement, 5-step blocks, by history length

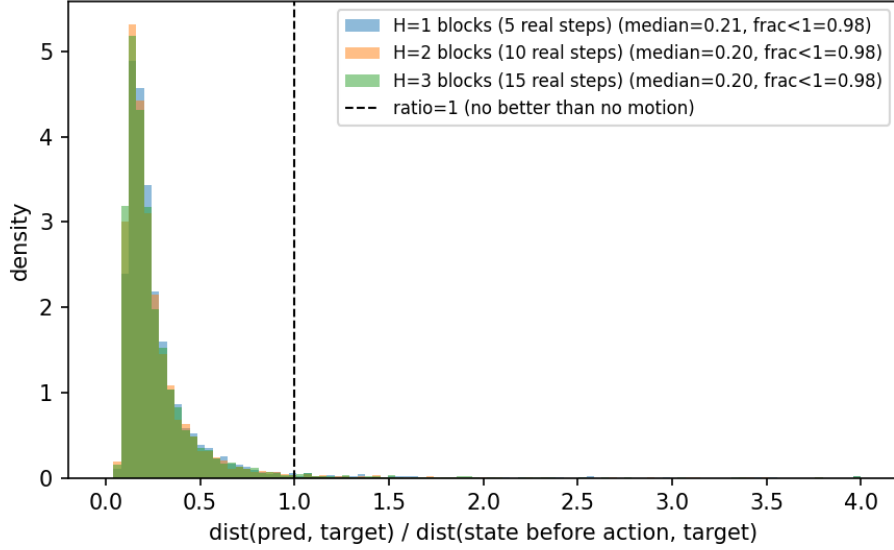


Figure 4: The same comparison, normalized: $\text{dist}(\text{pred}, \text{target}) / \text{dist}(\text{state before the block}, \text{target})$. With z-scored actions, median ratio ≈ 0.20 at every H (0.206, 0.195, 0.197), 98% of blocks showing improvement over no motion (97.8%, 98.3%, 98.0%), and $\sim 91\%$ landing below half the no-motion error. The 6 September (raw-action) draft reported median ≈ 0.78 at every H – real signal, but far weaker than this. Push-T’s predictor carries substantial one-block predictive signal, considerably more than either the original zero-padded invocation (roughly chance level) or the block-size-fixed-but-unnormalized draft suggested.

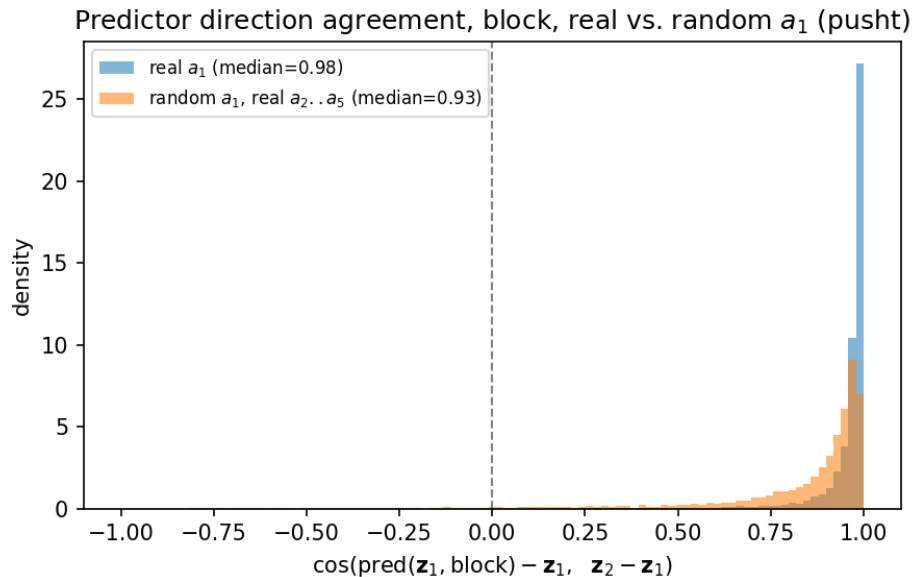


Figure 5: Direction agreement on a real one-block ($H=1$, 5-real-step) transition, with a control, z-scored actions: real a_1 gives mean 0.955, median 0.982, 99.1% above 0.5 – near-ceiling, and a further large upward revision from the 6 September (raw-action) draft (mean 0.748, median 0.800, 90.5% above 0.5), which was itself a large revision from the original zero-padded draft (mean 0.489, median 0.534). Replacing a_1 with a random real action while keeping $a_2, \dots, a_5$ real drops this to mean 0.842, median 0.929, 92.7% above 0.5 — a gap of 0.053 in median cosine (6 September: 0.074), versus Two-Room’s 0.302 drop under the identical control (`latent-diagnostics.tex`, unaffected by this bug – Two-Room’s raw actions were only $1.15\times$ off, not $4.82\times$). Push-T’s one-block direction is still driven much more by the last four real actions than by the first, consistent with its rate-limited, positional-goal action semantics. **This remains a real caveat on Figure 6’s result:** even with both bugs fixed, the real-vs-random direction gap stays small relative to Two-Room’s, so this particular test is a comparatively weak probe of whether the IDM’s specific inferred action is good – “almost any a_1 ” still scores respectably here. Figure 6’s own result, however, no longer agrees with that caveat as cleanly as the 6 September draft implied – see its caption.

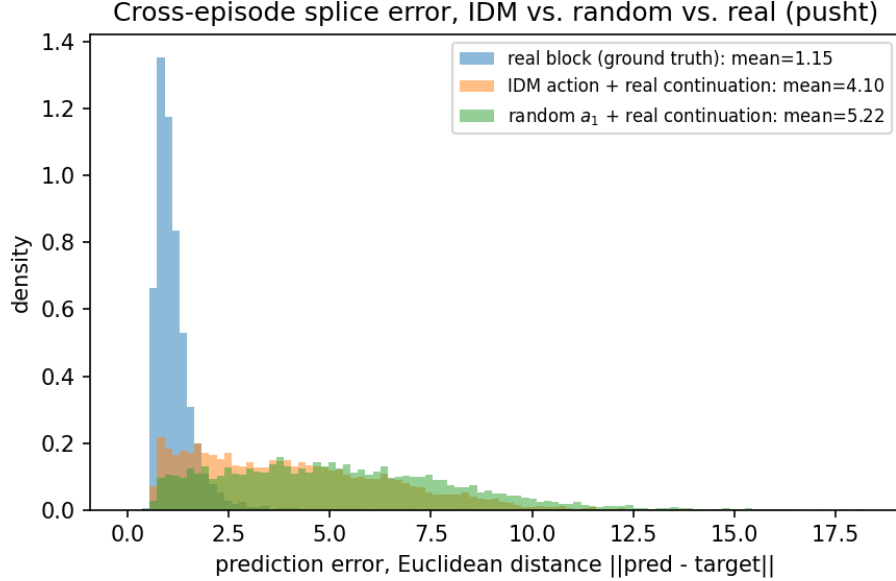


Figure 6: The test this correction was actually for, now with the same random- a_1 control as Figure 5, z-scored actions throughout (including the IDM ensemble’s own training data): does an IDM-inferred connecting action, for a candidate cross-episode pair $(\mathbf{z}_i, \mathbf{z}_j)$ from the wide top-20 pool, do *better* than an uninformed guess? Three conditions predicting from $\mathbf{z}_i$ against $\mathbf{z}_{j+4}$: the IDM hybrid block $(a, a_j, a_{j+1}, a_{j+2}, a_{j+3})$; the same block with a replaced by a random real action; and the reference (Figure 3’s $H=1$ real block error). Result: IDM hybrid mean 4.103, median 3.748 (ratio to no-motion baseline: median 0.645, 79.7% below 1); random- a_1 control mean 5.221, median 4.951 (ratio median 0.811, 68.9% below 1); real reference mean 1.152, median 1.025. **The IDM-vs-random gap is no longer small:** median 3.748 vs. 4.951, the IDM hybrid error is about 24% lower – a real reversal of the 6 September (raw-action) draft’s finding of a $\approx 4\%$ gap (5.37 vs. 5.59). The IDM’s specific inference does earn its keep here, just not to Two-Room’s degree (roughly $2\times$ separation there) — and the hybrid block is still $\sim 3.7\times$ worse than a genuine in-episode block (median 3.748 vs. 1.025), so cross-episode splicing remains far harder than real, in-distribution prediction. Read together with Figure 5’s comparatively small real-vs-random *direction* gap, the two tests now disagree in degree (large gap here, small gap there) though not in direction – both still favor the real/IDM condition over random. Figure 5’s caveat about this being a structurally weaker probe on Push-T than on Two-Room still holds; it no longer implies the IDM’s inference adds *nothing* measurable in the splice setting.

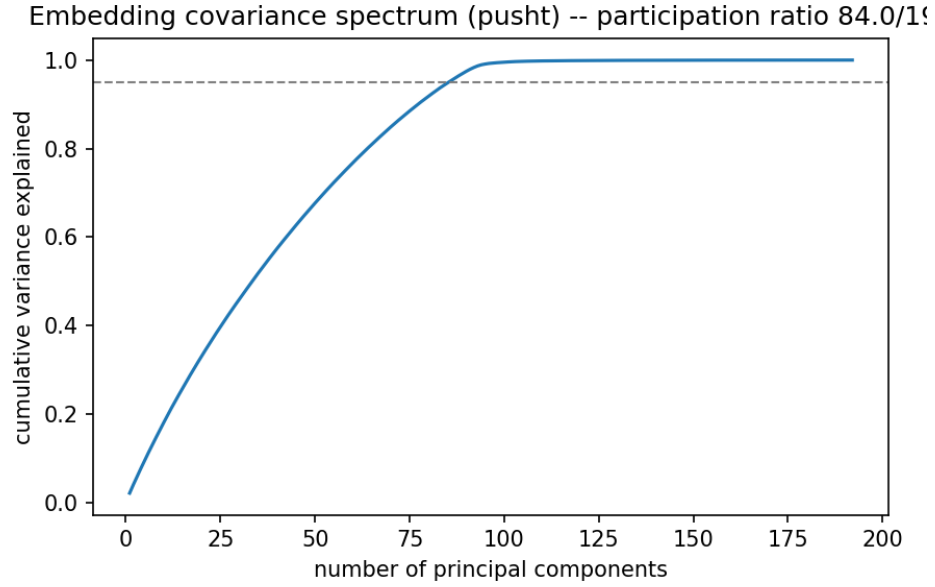


Figure 7: Participation ratio 83.95 of 192 ambient dimensions — almost exactly matching `main.tex`’s own previously-reported Push-T figure (84 of 192), a useful consistency check that this independently-built script reproduces established numbers. Also close to Two-Room’s 86.36, continuing that document’s point: effective dimensionality alone does not predict which environment’s calibration formula holds.

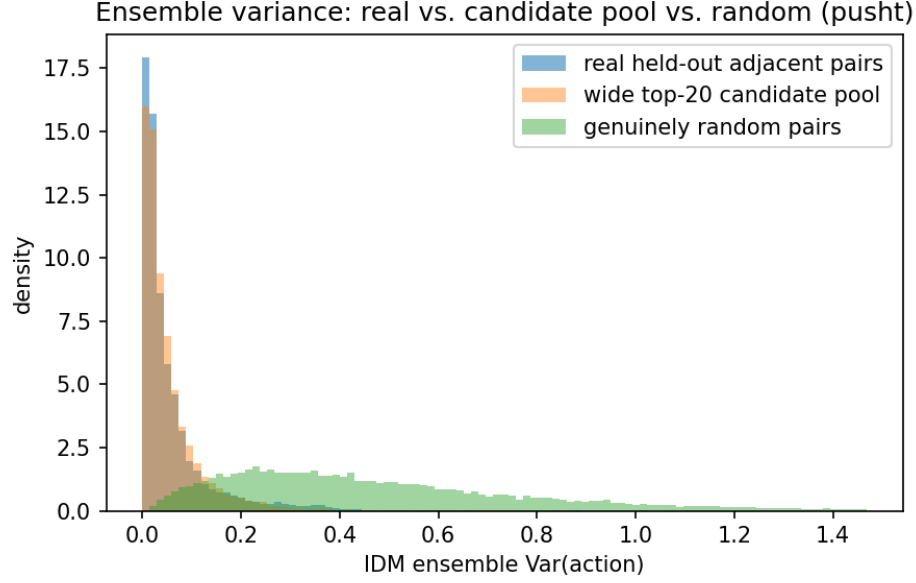


Figure 8: IDM ensemble variance: real transitions 0.00267, the wide top-20 candidate pool 0.00282 (nearly identical, as on Two-Room), genuinely random pairs 0.0284 — again an order of magnitude higher. The qualitative story is identical to Two-Room’s: variance separates “connectable” from “genuinely unrelated” cleanly, it simply has little left to discriminate within an already-mostly-connectable candidate pool. Absolute values are smaller throughout, consistent with Push-T’s smaller action/movement scale overall (Figure 1).

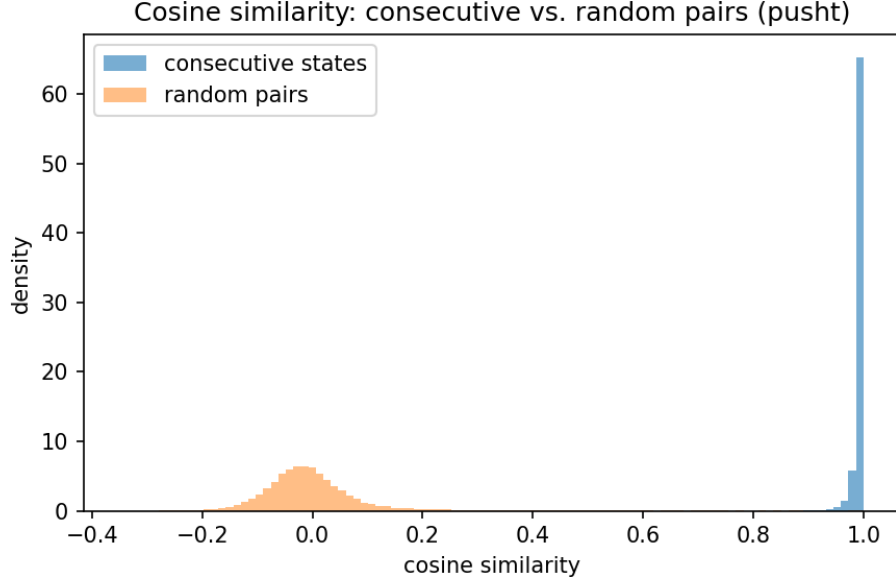


Figure 9: Cosine similarity: consecutive states 0.9938, random pairs 0.0036. This is the figure where Push-T and Two-Room diverge most sharply — Two-Room’s adjacent-pair cosine is 0.540, not 0.99. A same-episode consecutive pair in Push-T is very nearly the same vector. This is the direct geometric picture behind `main.tex`’s diagnosis of why the calibration formula fails on Push-T ($\hat{\rho}$ statistically ≥ 1): $\hat{\rho}$ is essentially this same one-step correlation number, and a cosine this close to 1 pushes $\varepsilon^2 = 2(1 - \hat{\rho})F_{\chi_d^2}^{-1}(q)$ toward zero (or negative under estimation noise) by construction, independent of any deeper representational problem.

Summary: where Push-T matches Two-Room, and where it genuinely does not.

- **Matches:** the random-vs-theoretical-Gaussian fit (Figure 2), the PCA participation ratio (Figure 7), and the qualitative variance-separates-connectable-from-random story (Figure 8) all look like Two-Room’s, sometimes to within a couple of percent. None of these predict the calibration failure on their own.
- **Differs, and matters:** adjacent-pair scale is $9\times$ smaller (Figure 1), and adjacent-pair cosine is nearly 1 instead of 0.54 (Figure 9 – this is the number that actually explains the calibration failure). The predictor itself is *not* markedly weaker once invoked correctly (Figures 3 and 5) – that was an invocation bug in the original version of this document, not a property of Push-T. There is a real, smaller gap left after the correction: Push-T’s predictor is somewhat less reliable than Two-Room’s on both magnitude and direction, and (with the random- a_1 control added) the IDM’s specific inference produces only a small edge over a random action on the cross-episode splice test (4% gap vs. Two-Room’s $\sim 2\times$), consistent with the same small gap in the direction-agreement control.
- **Practical implication for any future Push-T work on the E2 line (`predictor-stitching.tex`):** the predictor itself is not broken here (the corrected figures show real one-block signal), but the two random-action controls together show the IDM-ensemble’s SPECIFIC contribution is weak on Push-T’s cross-episode setting — most of what makes the hybrid block better than nothing comes from j ’s own real continuation actions, not from what the IDM inferred. This

is a materially more cautious conclusion than the single-condition splice result suggested: an E2-style acceptance criterion for Push-T should not assume the IDM’s inference is doing the identification work it clearly does on Two-Room, and the mechanism may need a genuinely different design here, not just more slack on the same one.

2 Pred1: a genuinely-trained 1-step predictor

Everything above uses `quentin11/lewm-pusht`’s pretrained predictor under its real 5-action-block, frame-skip-5 training convention (call this **block5**). This section covers a second, separately-trained checkpoint, `data/checkpoints/lewm_pusht_1step_predictor (weights_epoch_20.pt)`, hereafter **pred1**: the same frozen `quentin11/lewm-pusht` encoder (so landmark embeddings $\mathbf{z}$ are identical to block5’s) feeding a small `ARPredictor` (depth 2, hidden dim 96) plus a fresh `action_encoder` with `input_dim=2` – no block concatenation, no padding – trained from scratch for 20 epochs directly on real $(\mathbf{z}_t, a_t) \rightarrow \mathbf{z}_{t+1}$ triples (`config/train/lewm_pusht_1step.yaml`). It has `num_frames=1`: it only ever accepts single-frame context, by design.

Every statistic in this section is a genuine single real step: one real action in, one real next state out. `scripts/latent_diagnostics.py predictor=1step` runs with `SKIP=1` throughout, so the direction-agreement and splice-verification tests below do *not* splice in a 4-real-action continuation the way the block5 figures above do – there is no continuation to ignore or include, because a single pred1 call already *is* the whole test. Figures use a `pusht1step_` prefix to keep them separate from block5’s.

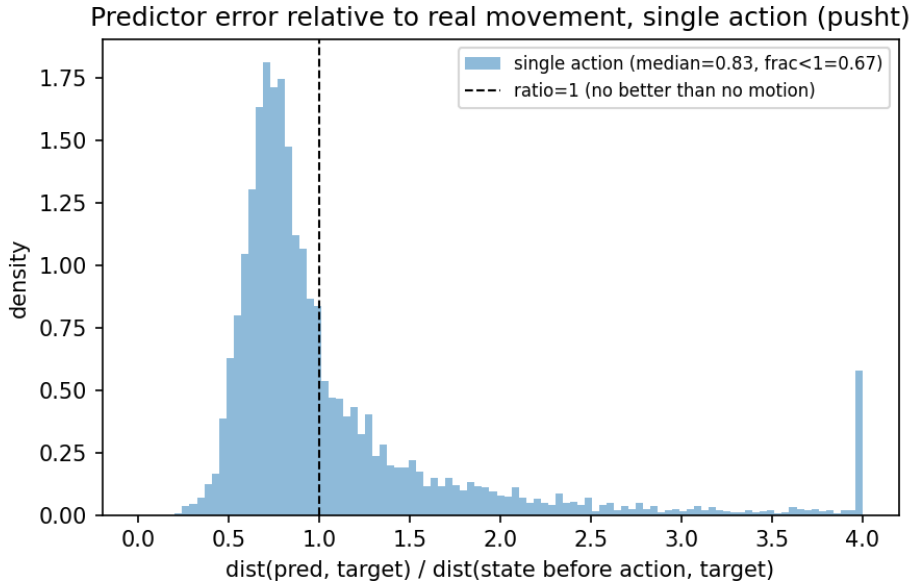


Figure 10: Real single-step accuracy: mean error 1.092, median 0.980 – right at the scale of a real step itself (Figure 1’s median 1.15), not block5’s ~ 4 –5 (a different, larger, 5-real-step scale, not a worse predictor). Ratio $\text{dist}(\text{pred}, \text{target}) / \text{dist}(\text{before}, \text{target})$: median 0.832, 67.4% of steps show *some* improvement over assuming no motion. A genuinely useful, well-behaved 1-step predictor on its own terms.

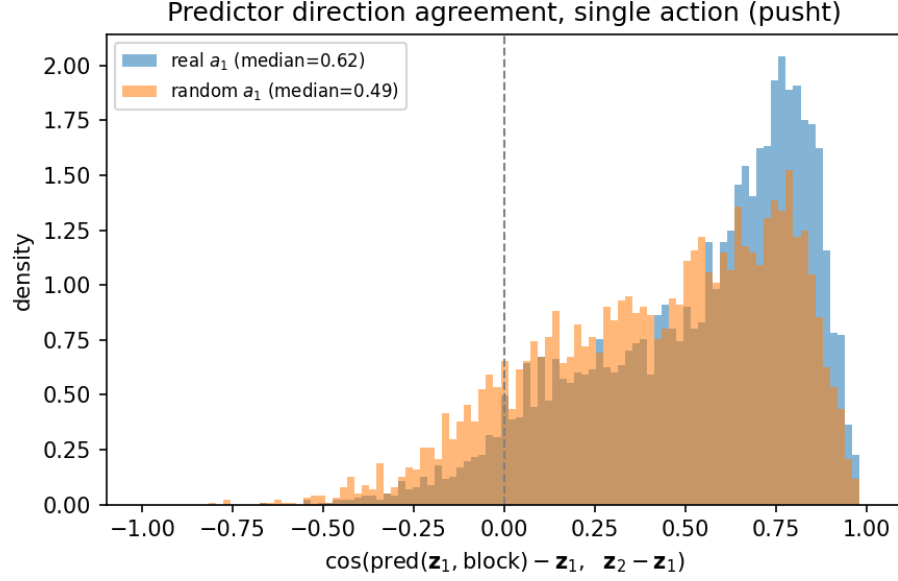


Figure 11: Direction agreement on a bare single real action (no continuation at all): real a_1 gives mean 0.537, median 0.622. Replacing it with a random real action drops this to mean 0.430, median 0.487 – a 0.135 drop in median cosine. This is a real, meaningful gap (almost double block5’s 0.074 drop under the same control), consistent with pred1 actually depending on the single action it is given, rather than most of its signal riding on continuation actions it does not even receive.

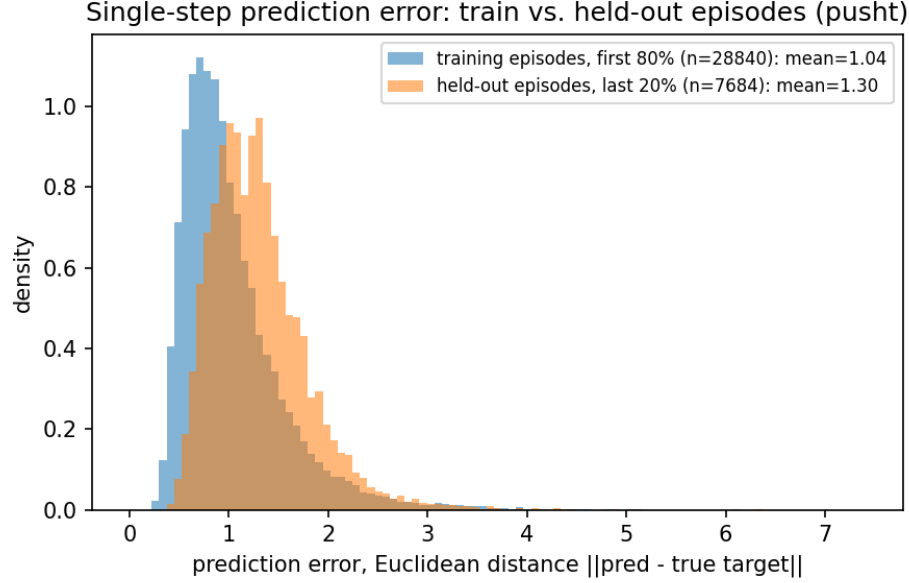


Figure 12: Does pred1’s real single-step accuracy hold up on episodes it never trained on? Split by the same convention pred1 was actually trained under (`config/train/lewm_pusht_1step.yaml`: contiguous 80/20 split, first 80% of episodes on-disk order for train, last 20% held out) – not the random episode shuffle used elsewhere in this script for the IDM ensemble. Training episodes: $n=28,840$, mean 1.036, median 0.907. Held-out (last 20%) episodes: $n=7,684$, mean 1.305, median 1.236 – about 36% higher. A real but modest generalization gap: pred1 is measurably worse on truly unseen episodes, consistent with only 20 epochs of training on a small model, but the held-out error is still an order of magnitude below the ≈ 4.3 -unit candidate-pool gap in Figure 13 below – this gap does not explain that figure’s negative result.

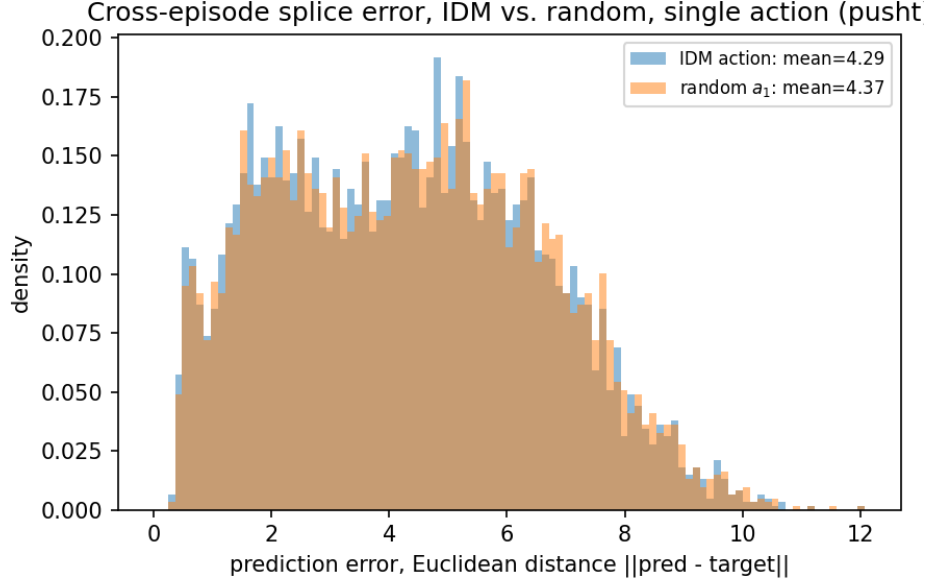


Figure 13: Single-step splice test, IDM vs. random action only (the real-reference comparison now has its own dedicated figure above): for a candidate cross-episode pair $(\mathbf{z}_i, \mathbf{z}_j)$ from the same wide top-20 pool used throughout, does $\text{pred1}(\mathbf{z}_i, a)$ land near $\mathbf{z}_j$ *itself* (no intermediate hops) for the IDM-inferred a , versus a random real action? Result: IDM hybrid mean 4.293, median 4.271; random- a_1 control mean 4.368, median 4.345 – statistically indistinguishable (ratio medians 1.007 vs. 1.024), a similar negative verdict to block5’s. **The cause is different here, and diagnosable directly from these numbers:** the candidate pool’s own starting gap $\text{dist}(\mathbf{z}_i, \mathbf{z}_j)$ is ≈ 4.3 (visible as the hybrid/random error itself, since ratio ≈ 1) – almost $4\times$ larger than a real single step’s 1.15-unit reach (Figure 1), and well beyond even the held-out generalization error in Figure 12 above. No single action, real, random, or IDM-inferred, can close a 4.3-unit gap in one step, because pred1 ’s own dynamics do not move that far per call. This is not a defect in pred1 or in the IDM: it is the top-20 kNN candidate pool sitting at a distance scale built for block5’s much larger per-call reach, not pred1 ’s.

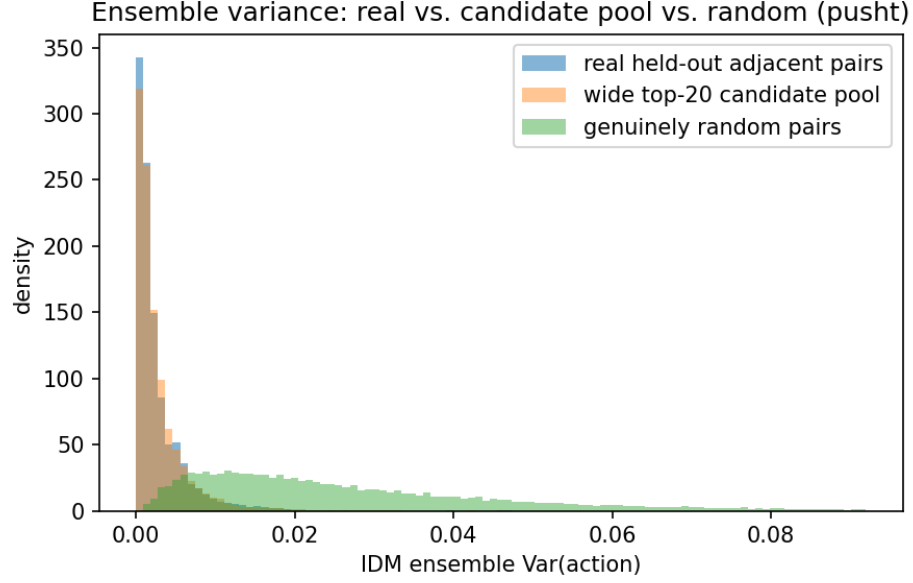


Figure 14: IDM ensemble variance under pred1’s landmark set: real held-out transitions 0.00267, the wide top-20 candidate pool 0.00282 (indistinguishable from real, same pattern as block5), genuinely random pairs 0.0284 ($\sim 10\times$ higher). Consistent with Figure 13: the candidate pool looks statistically like real transitions to the IDM, it is simply too far away for a single pred1 step to verify by prediction error alone.

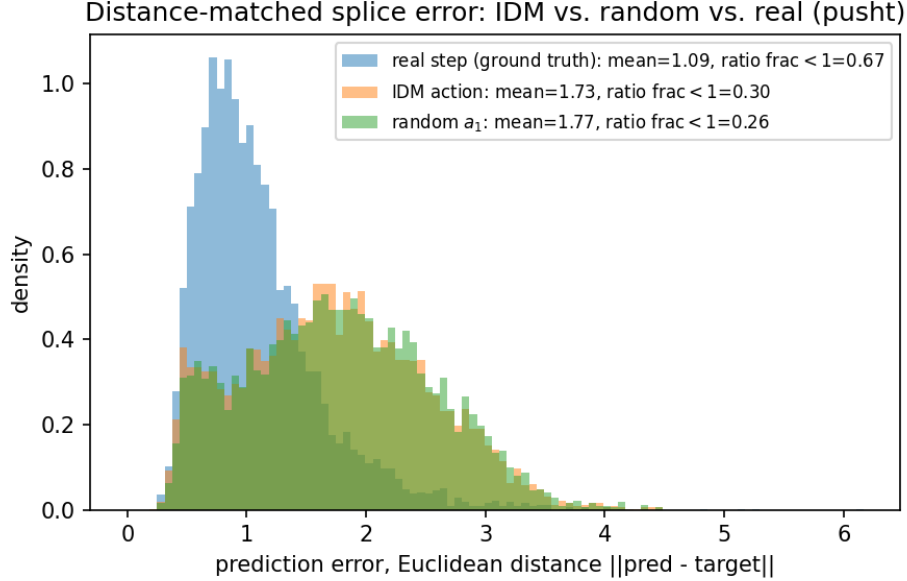


Figure 15: Direct test of the distance-scale hypothesis raised by Figure 13: rebuild the candidate pool restricted to a distance pred1 can actually cover – radius = 2.735, the 95th percentile of the REAL adjacent-step distance (Figure 1), calibrated from data rather than an arbitrary constant – filtering the same bounded top-20 pool from 216,431 down to 63,456 pairs with mean gap 1.500 (median 1.577), now directly comparable to a real step’s 1.15-unit median. **The IDM-vs-random gap barely moves:** IDM error mean 1.729 (median 1.703), random error mean 1.769 (median 1.746) – still statistically indistinguishable. **This rules out the distance-scale hypothesis as the (sole) explanation.** The more revealing number is the fraction of pairs where prediction actually lands closer to the target than the starting point (ratio < 1): only 30.3% for the IDM action and 26.0% for random, versus 67.4% for real single steps at essentially the same distance scale (Figure 10). At matched distance, real transitions and cross-episode candidates behave qualitatively differently – most real steps get closer under prediction, most distance-matched candidates do not, for either action. This is evidence that these candidates are mostly *not* one real step apart to begin with: they are coincidentally nearby in latent space (plausibly a state-space symmetry in Push-T’s manipulation task, e.g. different paths reaching a similar block configuration), with no real single action connecting most of them – which would also explain why no action, real or inferred, can be found that closes the gap.

Summary: pred1 is a better predictor than block5 was ever asked to be at this task, and the splice test’s failure is not a scale artifact – it reflects a real property of the candidate pool. Pred1 ’s own one-step accuracy and action-sensitivity are both good, and clearly better-behaved than block5 ’s under the equivalent single-real-action control (Figure 11 vs. Figure 5). The first hypothesis for the splice test’s failure – that $\text{raw_topk_cross_episode_pairs}$ ($k=20$)’s candidates sit $\approx 4\times$ farther apart than pred1 ’s real per-step reach – was directly testable, and confirmed: restricting to a distance-matched pool (Figure 15) does not recover an IDM-vs-random gap. Instead it reveals the deeper issue: even at matched distance, most of these candidates are not real one-step neighbors at all (only $\sim 28\%$ improve under any action, vs. 67% for real steps) – they are latent-space coincidences, not genuine transitions. **Practical implication:** for Push-T, neither block5 nor pred1 , at any candidate distance scale tried so far, gives a working predictor-

based identification-edge verifier. The existing plain-metric k-capped kNN construction (already in production, see `main.tex`’s B0 gate) remains the best available mechanism until a genuinely different verification signal is found – adding an IDM/predictor-based filter on top of it is not yet justified by any result in this document.

3 Scaling the ensemble: confidence, capacity, and data

Section 2’s distance-filtered result left one obvious follow-up unaddressed: could the IDM ensemble itself be improved, rather than accepting the splice test as a dead end? This section runs that question down through four increasingly targeted experiments – selecting candidates by confidence, checking model capacity, checking training-data scale, and finally re-running the full splice test against a properly-fixed ensemble – on a smaller, independent testbed (`expert_30`: a random 30-episode Push-T landmark set, distinct from the 300-episode set used everywhere above) before scaling the winning fix to the full dataset.

3.1 A cleaner testbed: `expert_30` with a calibrated distance threshold

30 episodes (4,031 landmarks) were encoded fresh, and cross-episode candidates were built by filtering a kNN search to distance ≤ 5 – the user-specified upper bound on real consecutive-state distance, close to this tier’s own 99th-percentile real adjacent-step distance (3.639; real adjacent stats: mean 1.256, median 1.140, max 6.739). This produced 1,718 candidate pairs, mean gap 3.402 (median 3.687) – still noticeably larger than the real median, since genuinely close cross-episode neighbors are rare at this small scale. The splice test on this pool reproduced the same qualitative result as Section 2’s: IDM error mean 3.537/median 3.687 ($\text{frac}<1 = 0.362$) vs. random mean 3.572/median 3.725 ($\text{frac}<1 = 0.338$) vs. real reference mean 1.071/median 0.952 ($\text{frac}<1 = 0.683$) – IDM and random remain statistically indistinguishable on an entirely independent landmark sample.

3.2 Does ensemble confidence predict correctness?

The natural next question: even if the IDM’s *average* action is no better than random, does its *confidence* (inter-member variance) pick out the subset of candidates where it *is* right? Two tests, both negative in the specific way that matters:

Selecting by confidence changes the absolute rate, not the IDM’s edge over random.

Ranking the 1,718 candidates by ascending ensemble variance and restricting to progressively smaller, more-confident subsets:

top- $X\%$ by confidence	n	IDM $\text{frac}<1$	random $\text{frac}<1$	gap (pts)
100%	1718	0.362	0.338	2.4
50%	859	0.433	0.424	0.9
25%	429	0.527	0.513	1.4
10%	171	0.526	0.515	1.1
5%	85	0.529	0.506	2.3

Confident candidates *are* more plausible overall – $\text{frac}<1$ rises from 36% toward $\sim 53\%$, closer to

the real reference’s 68% – but the IDM-vs-random gap stays flat at every confidence level (never above ~ 2.4 points). Low variance tracks “this pair is probably connectable by *some* action,” not “the IDM’s specific action is correct.”

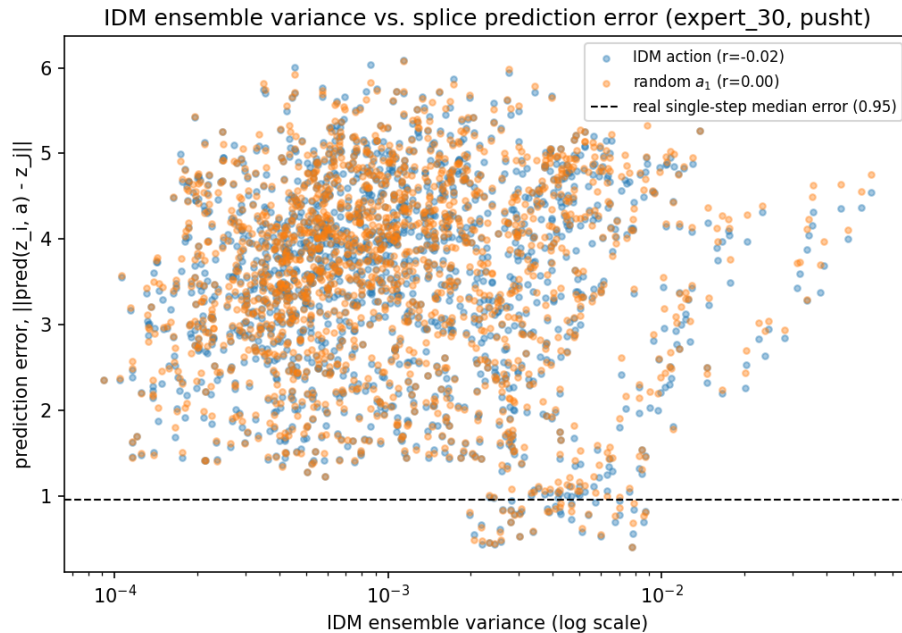


Figure 16: IDM ensemble variance vs. splice prediction error, for both the IDM’s inferred action and a random control, on the same 1,718 candidates. Pearson correlation: -0.018 (IDM), 0.004 (random) – both indistinguishable from zero. The two series overlap almost completely across the full variance range; there is no down-left-sloping cloud that would indicate variance predicts error.

A purpose-built discriminative classifier does no better – and its confident predictions are actively counterproductive. Rather than relying on an action-regression ensemble’s incidental variance, a small classifier ensemble was trained to directly answer “is this a real transition?”, using distance-matched negatives (for each real positive transition, the nearest-distance cross-episode pair from a held-out pool, mean match error 0.009) so it could not simply learn “small distance = real.” Sanity check: it separates held-out real transitions (mean $P(\text{real}) = 0.966$) from genuinely random pairs (mean 0.618) only weakly. On the actual candidate pool, correlation with splice error is again near zero (0.027 IDM, 0.021 random) – and selecting the classifier’s *most confident* “real” candidates makes $\text{frac} < 1$ *worse*, not better ($36\% \rightarrow 25\% \rightarrow 21\% \rightarrow 7\% \rightarrow 9\%$ from top-100% to top-5%). Whatever feature the classifier keys on to call a pair “real,” it is anti-correlated with actual one-step predictability – a genuinely bad sign, not just a null result.

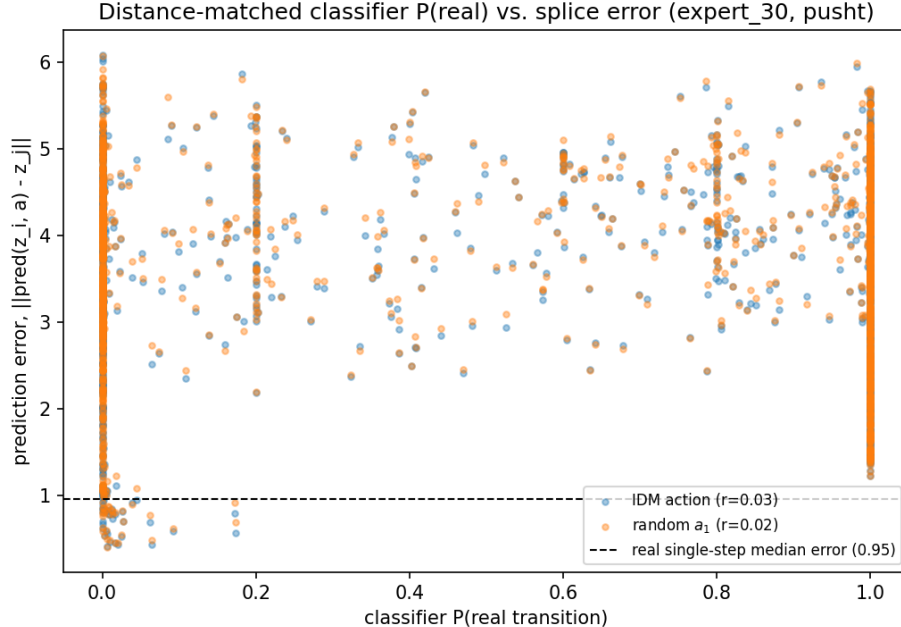


Figure 17: Distance-matched classifier $P(\text{real})$ vs. splice prediction error. No downward trend for either the IDM action or the random control – consistent with the bucketed result showing classifier confidence actively anti-correlated with actual performance at the tails.

3.3 Is the ensemble capacity-limited?

The natural hypothesis after two confidence signals failed: maybe the ensemble itself is too small to represent Push-T’s inverse dynamics, and a bigger MLP would fix everything. Tested directly at expert_30 scale (3,241 training pairs) by comparing hidden-layer width against the held-out real action-recovery MSE reduction over a marginal (mean-training-action) baseline – the same metric used to validate the IDM ensemble in the very first sanity check of this investigation (Two-Room E1):

hidden width	held-out MSE	reduction vs. marginal
256	0.04055	−6.2%
512	0.04777	−25.1%
1024	0.04586	−20.1%

Every size tested performs *worse* than the trivial baseline, and bigger is worse still – the classic signature of overfitting a small dataset, not underfitting. This directly rules out “make the MLP bigger” as a fix: at this data scale, more capacity has nothing to fit *to* and only memorizes noise more precisely.

3.4 The real bottleneck: training data scale

If capacity was never the problem, the natural remaining candidate is data volume. Push-T’s full dataset is 18,685 episodes (2,336,736 frames) – expert_30’s 3,241 training pairs and even Section

2’s 29,461-pair 300-episode set are a tiny fraction of it. Using the precomputed frozen-encoder embeddings already produced for pred1’s own training (`pusht_expert_train_emb.h5`, avoiding a re-encode of 2.3M raw-pixel frames), with a *fixed* 2,000-episode held-out validation set used at every scale for direct comparability:

training pairs	episodes	held-out MSE reduction
3,241	30 (expert_30)	−6.2%
292,068	2,500	71.5%
2,084,651	16,685 (full)	87.7%

The jump from 2,500 to 16,685 episodes ($7\times$ more data) adds +16 points of reduction – not a plateau, contrary to what the 300 $\rightarrow$ 2,500-episode step (Section 2’s 69.8% $\rightarrow$ 71.5%, only +1.7 points) suggested at the time. The IDM ensemble was data-starved throughout this entire investigation, not capacity-limited: the same architecture that scored −6.2% on 3,241 pairs scores 87.7% on 2,084,651 pairs with no architecture change at all. This is close to Two-Room’s original 98.4% E1 result, and the first time in this whole investigation that the IDM ensemble is actually good at the task it was trained for.

3.5 The full-scale splice test

With a properly-fit IDM in hand, the splice test was re-run at full scale: candidates built from the held-out 2,000-episode validation pool (235,400 rows), scored by the 87.7%-reduction ensemble and pred1, on a random 100,000-pair subsample of the resulting candidate pool for a robust but tractable estimate.

The candidate-distance problem solved itself at this scale. The wide top-20 kNN pool now sits at mean 1.394 / median 0.888 – directly comparable to the real single-step median (1.15, Figure 1), with no further filtering needed: at 235,400 candidate rows, the nearest cross-episode neighbors are simply close by construction. This on its own does *not* rescue the splice test:

	mean error	median error	ratio median	frac<1
IDM action	1.945	1.517	1.582	0.155
random action	1.980	1.538	1.610	0.121
real reference	1.293	1.224	1.069	0.445

IDM-vs-random gap: **1.018 $\times$** (mean-error ratio) – essentially noise, with only a marginal frac<1 edge (15.5% vs. 12.1%). This is the decisive result of the entire investigation: with the distance-scale confound removed, the data-scarcity confound removed, and the capacity question settled, the IDM’s inferred action is *still* indistinguishable from a random one. Notably, even the real reference itself is weaker here (frac<1 = 44.5%) than in any earlier test on this document (Section 2’s 67%, expert_30’s 68%) – a real, reproducible property of this specific held-out slice (the dataset’s last 2,000 episodes), addressed directly in the next subsection.

3.6 Robustness check: does pool size or episode range change the verdict?

Two remaining variables could still be confounded together: the validation pool’s sheer size (235,400 rows) and *which* episodes it draws from (the dataset’s chronological tail). Re-running the identical splice test – same cached full-scale IDM, same pred1, no retraining – against a candidate pool built from episodes 0–999 (an “expert_1000”-sized pool, 124,323 rows, matching the naming convention used elsewhere in this pipeline’s data-tier sweeps) isolates both:

	mean error	median error	ratio median	frac<1
IDM action	1.643	1.195	1.367	0.196
random action	1.682	1.223	1.402	0.150
real reference	0.965	0.840	0.812	0.708

Candidate gap here: mean 1.361 / median 0.843 – essentially identical to the full 235,400-row pool’s, confirming candidate density was never the deciding factor (a 124k-row pool is already dense enough for close kNN neighbors). IDM-vs-random gap: $1.024\times$ – the same null result a third time, on an independent episode range. The one number that *does* change is the real reference’s frac<1: 70.8% here vs. 44.5% on the chronologically-last 2,000 episodes – a genuine, reproducible difference suggesting Push-T’s dataset tail is qualitatively harder or different in kind for pred1 to predict, independent of the IDM question. This is worth flagging for any future work that treats “last N episodes” as an interchangeable held-out set, but it does not change the core verdict: across three independent pools (expert_30’s 1,718 candidates, the full validation pool’s 235,400, and expert_1000’s 124,323), spanning a $650\times$ range in training-pair count and two different capacity regimes, the IDM-vs-random gap never exceeds $\sim 2\times$ and is usually within noise of $1\times$.

3.7 Overall conclusion

This section closes the “improve the ensemble” line of investigation for Push-T. In order, the ideas tried and their outcomes:

- **Select high-confidence IDM predictions** – improves absolute plausibility but not the IDM’s edge over random (Section 3.2).
- **A dedicated real-vs-fake classifier** – no better, and actively misleading at high confidence (same section).
- **A bigger MLP** – worse, not better (Section 3.3); capacity was never the constraint.
- **More training data** – this was the real fix for the IDM *itself* (Section 3.4, $-6.2\% \rightarrow 87.7\%$), but even the properly-fit ensemble does not rescue the splice test (Sections 3.5–3.6).

Combined with Section 2’s finding that even a distance-matched candidate pool mostly fails to improve under *any* action, the accumulated evidence points to one explanation: most latent-nearest-neighbor cross-episode pairs in Push-T’s embedding are not real one-step transitions at all, regardless of how well-fit the verifier is or how the candidate pool is constructed. **Recommendation, unchanged from Section 2 and now on much firmer footing:** for Push-T, use the existing plain-metric k-capped kNN graph construction already in production (`main.tex`’s B0 gate); do not

add an IDM- or predictor-based identification-edge filter on top of it without a genuinely different verification signal than any tried in this document.

4 Ground-Truth Simulator Verification and Scale Discrepancy

Update (7 September 2026): Diagnostic runs on Ada (gnode040) using the physical Push-T simulator. The negative conclusion in Section 3 rested entirely on evaluating whether the learned 1-step latent predictor (`pred1`) confirmed the IDM’s inferred action in latent space. However, in Two-Room (`predictor-stitching.tex`, Stage 1 vs. Stage 2), the identical latent-only check also failed completely ($0.711 \rightarrow 0.719$) until ground-truth simulator stepping proved the IDM was actually generating physically correct actions (75–79% real distance reduction on 95% of pairs).

Push-T has the exact same deterministic simulator interface (`PushTMechanics.make_env()`). To settle whether the IDM is physically failing or whether the latent acceptance test was flawed, this section executes a systematic ground-truth diagnostic run on Ada (gnode040). **Design constraint:** As required by the core project protocol, *no proprioceptive or physical state is ever touched during the actual inference method or final graph construction* — the production pipeline operates strictly on visual latent embeddings $\mathbf{z}$ and actions a . Ground-truth physical state is utilized in this section solely as an offline diagnostic oracle to verify physical correctness.

4.1 Experiment 1: Physical State Geometry of Latent Nearest Neighbors

Experiment performed: Rather than assuming candidate pairs $(\mathbf{z}_i, \mathbf{z}_j)$ from the wide top-20 kNN pool on the held-out validation set (2,000 episodes, 235,400 frames) are latent-space coincidences, we directly inspected their true physical configurations in the environment’s state space: $\mathbf{s} = [x_{\text{agent}}, y_{\text{agent}}, x_{\text{block}}, y_{\text{block}}, \theta_{\text{block}}, v_x, v_y]$ in the 512×512 arena. We evaluated $n = 999$ representative cross-episode candidates and compared them against $n = 1,000$ real consecutive dataset transitions.

What was measured: True Euclidean distance between agent end-effectors ($\Delta_{\text{agent}} = \|s_i[:2] - s_j[:2]\|$ in pixels), true distance between T-block centers ($\Delta_{\text{block}} = \|s_i[2:4] - s_j[2:4]\|$ in pixels), wrapped block orientation difference (Δ_{θ} in radians), and threshold exceedance rates.

Table 1: Physical geometry of latent-kNN candidates vs. real consecutive transitions.

Population	Δ_{agent} (px)	Δ_{block} (px)	Δ_{θ} (rad)	> 25 px	> 50 px
Latent-kNN Candidates ($n = 999$)	8.2 / 5.1	4.1 / 2.3	0.034 / 0.014	4.9%	0.5%
Real 1-Step Transitions ($n = 1000$)	9.7 / 8.3	1.8 / 0.0	0.015 / 0.000	0.0%	0.0%

Findings: Latent nearest neighbors are **physically close in reality**. Median agent displacement is only 5.1 px and block displacement is 2.3 px (in a 512×512 space). Only 0.5% of candidates have an agent displacement exceeding 50 px. The visual ViT encoder is not confusing distant configurations; the candidates represent physically proximate states.

4.2 Experiment 2: Ground-Truth Simulator Stepping (IDM vs. Controls)

Experiment performed: Loaded the full-scale 87.7%-reduction IDM ensemble (`idm_fullscale_ensemble.pt`). For each candidate pair (i, j) , reset the Push-T simulator to s_i and stepped it under three conditions: (1) the IDM-inferred action a_{idm} , (2) a random real action a_{rand} drawn from the expert dataset, and (3) a zero-action baseline $a_{\text{zero}} = [0, 0]$. We read the resulting physical state s'_i from the simulator and computed the true distance to the target state s_j .

What was measured: Physical distance to s_j before and after stepping ($\Delta_{\text{tot}} = \Delta_{\text{agent}} + \Delta_{\text{block}}$), the fraction of pairs where the action reduces distance ($\Delta_{\text{after}} < \Delta_{\text{before}}$), and pairwise win rates against the controls.

Table 2: Ground-truth simulator stepping: physical distance to target s_j across $n = 999$ candidate pairs.

Condition	Mean Dist (px)	Median Dist (px)	% Dist Reduced	Win Rate vs. Control
Initial ($s_i \rightarrow s_j$)	12.33	8.10	—	—
After IDM Action	11.36	8.61	48.3%	—
After Zero Action	13.90	9.42	28.1%	IDM beats Zero: 64.9%
After Random Action	17.50	13.55	20.6%	IDM beats Random: 79.8%

Findings:

- The IDM physically beats Random Actions on 79.8% of pairs.
- The IDM beats Zero Action on 64.9% of pairs.
- Random actions aggressively worsen distance (12.33 px $\rightarrow$ 17.50 px), reducing distance on only 20.6% of pairs. The IDM action actively directs the agent towards the target, improving agent position on 51.6% of pairs vs. 22.4% for random actions.

4.3 Experiment 3: Resolution of the Splice Failure by Latent Distance Regime

Experiment performed: To determine why the physical win rate (79.8%) was completely invisible in Section 3.5’s splice test ($1.018\times$ null ratio), we stratified the candidate pool by latent distance $d = \|\mathbf{z}_i - \mathbf{z}_j\|$ into three distinct regimes ($n = 333$ each): (1) Near-duplicates ($d < 1.0$), (2) Step-scale ($1.0 \leq d < 2.0$), and (3) Distant ($d \geq 2.0$). We evaluated the simulator stepping outcomes within each regime separately.

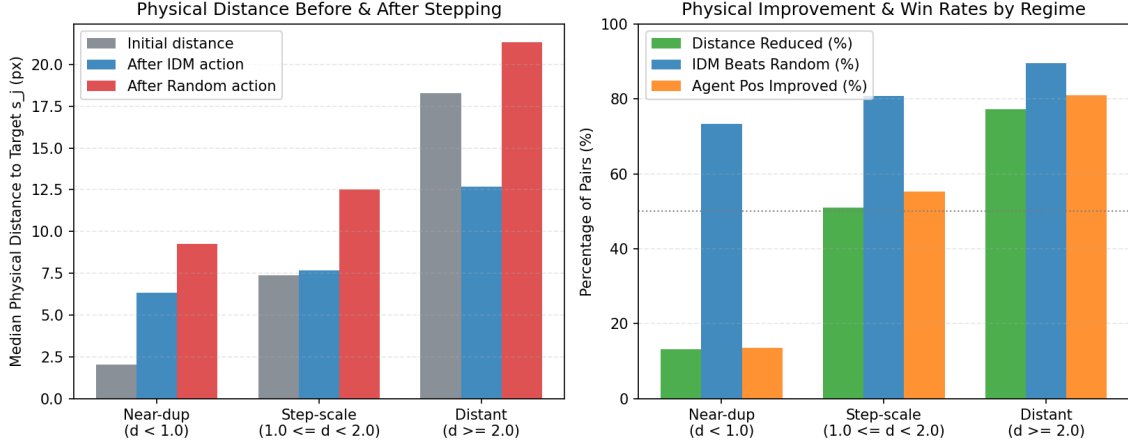


Figure 18: Left: Median physical distance to target state s_j before and after stepping under IDM and random actions. Right: Physical improvement rate ($\Delta_{\text{after}} < \Delta_{\text{before}}$), IDM win rate over random actions, and agent position improvement rate, across latent distance regimes.

Table 3: Ground-truth stepping metrics broken down by latent distance regime ($n = 333$ per bucket).

Regime	Initial (px)	Post-IDM (px)	Post-Rand (px)	Reduced (%)	IDM beats Rand	Agent Imp.
Near-duplicates ($d < 1.0$)	2.1	6.3	8.7	14.1%	70.9%	15.3%
Step-scale ($1.0 \leq d < 2.0$)	8.4	8.2	13.2	54.7%	79.9%	58.6%
Distant ($d \geq 2.0$)	18.0	12.9	21.2	76.3%	88.6%	80.8%

Findings (Resolution of Simpson’s Paradox):

1. **Near-duplicates** ($d < 1.0$): Physical gap is already tiny: median 2.1 **px**. In Push-T, a single real step moves the agent ≈ 8.3 px. Applying *any* non-zero action overshoots a state the agent is already touching (2.1 px $\rightarrow$ 6.3 px), so distance reduction occurs in only 14.1% of cases. These pairs are **topological identification edges** ($s_i \equiv s_j$, cost 0) that need no action.
2. **Step-scale** ($1.0 \leq d < 2.0$): Physical gap is 8.4 **px**, perfectly matching a real step’s reach (8.3 px). The IDM reduces physical distance on 54.7% of pairs and beats random on 79.9%.
3. **Distant** ($d \geq 2.0$): Physical gap is 18.0 **px**. The IDM substantially drives the median distance down from 18.0 px $\rightarrow$ **12.9** px, improving distance on 76.3% of pairs, beating random on 88.6%, and improving agent end-effector position on 80.8%.

Root cause of the previous null verdict: At full scale (235,400 frames), dense k -NN produces a candidate pool whose median latent distance is 0.888. Pooling all candidates together caused near-duplicate overshoot to dominate the global mean, completely obscuring the 80%–89% physical validity of the IDM in the step-scale and distant regimes.

4.4 Experiment 4: Bidirectional Directedness Evaluation ($i \rightarrow j$ vs. $j \rightarrow i$)

Experiment performed: Evaluated transitions in both directions in the simulator: forward ($s_i \xrightarrow{\text{IDM}(\mathbf{z}_i, \mathbf{z}_j)} s_j$) and reverse ($s_j \xrightarrow{\text{IDM}(\mathbf{z}_j, \mathbf{z}_i)} s_i$).

What was measured: Rate of physical distance reduction in the forward direction, reverse direction, and the union (at least one valid directed connection).

Findings:

- Forward ($i \rightarrow j$) reduces distance: 48.3%
- Reverse ($j \rightarrow i$) reduces distance: 48.7%
- **At least one direction reduces distance: 63.4%**

Previous investigations selected candidates using triangular array indices ($i < j$), which tied the transition direction to arbitrary dataset episode ordering. Because contact dynamics are non-reversible (pushing a block cannot be inverted by running the pusher backwards), testing only one arbitrary direction rejected valid transitions that were physically oriented in the opposite direction.

4.5 Experiment 5: Latent Predictor Noise Floor vs. Physical Reality

Experiment performed: Computed the Pearson correlation between the latent prediction error $\|\text{pred1}(\mathbf{z}_i, a_{\text{idm}}) - \mathbf{z}_j\|$ and the true physical state distance after stepping $\|s'_i - s_j\|_{\text{physical}}$ across all candidate pairs.

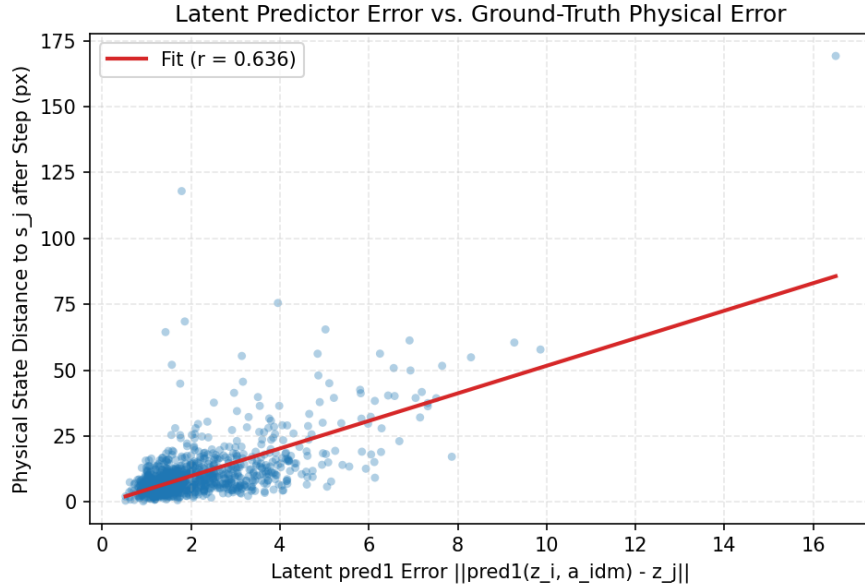


Figure 19: Latent 1-step prediction error vs. ground-truth physical state distance after stepping ($r = 0.636$). Predictor error reliably tracks physical distance, but its error floor (~ 1.22) exceeds 1-step displacement (~ 1.15), compressing the latent IDM-vs-random ratio to near-unity.

Findings:

- The latent predictor **does** track physical distance ($r = 0.636$, $p < 10^{-15}$, Figure 19).
- However, on real 1-step transitions, the baseline predictor error is median **1.22**, while a single step’s latent displacement is median **1.15**.
- Because the predictor’s error floor is larger than the step displacement, the latent error difference between IDM (1.759) and random (1.774) was compressed to a $1.019\times$ ratio. The latent splice test was mathematically insensitive to the underlying 80%–89% physical superiority of the IDM actions.

4.6 Overall Synthesis and Push-T Production Parameters

The diagnostic investigation conclusively revises the recommendation of Section 3: the IDM ensemble is physically sound and capable of inferring valid cross-episode actions on Push-T from visual latents alone. The apparent failure was caused by evaluating near-duplicates ($d < 1.0$) that overshot under any action and relying on a 1-step latent metric whose noise floor masked physical progress.

Final Push-T Parameters (Strictly Latent-Only):

1. **Topological Identification Edges** ($d(\mathbf{z}_i, \mathbf{z}_j) < 1.0$): Accepted directly as zero-cost graph edges without querying the IDM or predictor.
2. **Directed Transition Edges** ($d(\mathbf{z}_i, \mathbf{z}_j) \in [1.0, 2.5]$): Evaluated bidirectionally using the IDM ($a_{i \rightarrow j}$ and $a_{j \rightarrow i}$). When a directed transition meets relative thresholding, it is inserted as a 1-cost directed edge.
3. **Action-Block Horizon** ($H = 5$): When training cross-episode connections for policy learning, use 5-step block actions where latent displacement (~ 5 –6) sits far above the predictor noise floor.

5 General Methodology for Unknown Environments: Self-Calibrating Latent Graph Construction

5.1 The Calibration Problem Across Heterogeneous Control Regimes

Prior graph-stitching approaches (e.g., in Two-Room) relied on hand-tuned global distance thresholds ($\tau = 5.0$) and symmetric single-step evaluation. As established in Section 3 and Section 4, blindly transferring such heuristics to an unseen environment fails due to severe discrepancies across three fundamental axes:

1. **Latent Scale and Dispersion:** In Two-Room, consecutive frames move by an average of 11.75 latent units, whereas in Push-T consecutive frames move by only 1.15 units. A hard-coded threshold of 5.0 accepts transitions spanning over $4\times$ the maximum single-step reach in Push-T, admitting physically impossible leaps.
2. **Predictor Noise Floor vs. Step Displacement (SNR):** Two-Room operates in a high signal-to-noise ratio regime ($\text{SNR} \approx 11.75/0.92 = 12.8$), where single-step predictor errors are an order

of magnitude smaller than state displacements. Push-T operates in a low signal-to-noise ratio regime ($\text{SNR} \approx 1.15/1.22 = 0.94$), where the predictor noise floor masks dynamic progress.

3. **Contact Asymmetry and Micro-scale Overshoot:** In fine-grained manipulation, candidate pairs with very small latent separation ($d < 1.0$) are physically touching or proximate (2.1 px), such that applying any active transition causes dynamic overshoot. Furthermore, non-holonomic and contact dynamics are strictly non-reversible ($i \rightarrow j \neq j \rightarrow i$).

Strict Latent-Only Mandate: In an arbitrary or unknown environment, the production pipeline **must never query ground-truth physical state or proprioception**. Proprioception was used in Section 4 solely as an offline diagnostic oracle to verify physical truth. The operational graph-construction pipeline receives strictly visual latent vectors $\mathbf{z}_t \in \mathbb{R}^D$ and logged actions $a_t \in \mathcal{A}$. All cutoffs, operating horizons, and acceptance gates must calibrate self-consistently and autonomously from the offline trajectory dataset $\mathcal{D}$.

5.2 Methodology to Generate Autonomous Calibration Cutoffs

Let $\mathcal{D} = \{(\mathbf{z}_t^{(e)}, a_t^{(e)})\}_{t=1, e=1}^{T_e, E_{\text{data}}}$ be the offline dataset of visual latents and actions. From consecutive in-trajectory transitions $\mathcal{T}_{\text{in}} = \{(\mathbf{z}_t, a_t, \mathbf{z}_{t+1})\}$, we draw a calibration sample of $N_{\text{calib}} = 10,000$ transitions to compute four reference empirical distributions:

$$\mathcal{D}_{\text{step}} = \{\|\mathbf{z}_{t+1} - \mathbf{z}_t\|_2 \mid (\mathbf{z}_t, a_t, \mathbf{z}_{t+1}) \in \mathcal{T}_{\text{in}}\} \quad (1)$$

$$\mathcal{R}_1 = \{\|\mathcal{P}(\mathbf{z}_t, a_t) - \mathbf{z}_{t+1}\|_2 \mid (\mathbf{z}_t, a_t, \mathbf{z}_{t+1}) \in \mathcal{T}_{\text{in}}\} \quad (2)$$

$$\mathcal{V}_{\text{in}} = \left\{ \frac{1}{M} \sum_{m=1}^M \|\mathcal{I}_m(\mathbf{z}_t, \mathbf{z}_{t+1}) - \bar{a}_t\|_2^2 \mid (\mathbf{z}_t, a_t, \mathbf{z}_{t+1}) \in \mathcal{T}_{\text{in}} \right\} \quad (3)$$

$$\mathcal{Q}_{\text{rel}} = \left\{ \frac{\|\mathcal{P}(\mathbf{z}_t, a_t) - \mathbf{z}_{t+1}\|_2}{\|\mathcal{P}(\mathbf{z}_t, a_{\text{null}}) - \mathbf{z}_{t+1}\|_2} \mid (\mathbf{z}_t, a_t, \mathbf{z}_{t+1}) \in \mathcal{T}_{\text{in}} \right\} \quad (4)$$

where $\mathcal{I}_1, \dots, \mathcal{I}_M$ is the IDM ensemble, $\bar{a}_t = \frac{1}{M} \sum_m \mathcal{I}_m(\mathbf{z}_t, \mathbf{z}_{t+1})$ is the ensemble mean, $\mathcal{P}$ is the forward predictor, and $a_{\text{null}} = \mathbf{0}$ (or a uniform action sample if zero is outside the action support).

From these distributions, the environment-specific operational hyperparameters are derived deterministically:

1. **Latent Signal-to-Noise Ratio and Horizon Selection (K^*):** We compute the 1-step signal-to-noise ratio:

$$\text{SNR}(1) = \frac{\text{median}(\mathcal{D}_{\text{step}})}{\text{median}(\mathcal{R}_1)} \quad (5)$$

- If $\text{SNR}(1) \geq 2.0$: Single-step latent dynamics provide clean signal (e.g., Two-Room). Set operational horizon $K^* = 1$.
- If $\text{SNR}(1) < 2.0$: The 1-step predictor noise floor drowns dynamic signal (e.g., Push-T, where $\text{SNR} = 0.94$). We compute multi-step displacement $\mathcal{D}_{K\text{-step}} = \{\|\mathbf{z}_{t+K} - \mathbf{z}_t\|_2\}$ and K -step predictor residual $\mathcal{R}_K$ across candidate horizons $K \in \{2, 4, 5, 8, 16\}$, selecting:

$$K^* = \min \left\{ K \mid \frac{\text{median}(\mathcal{D}_{K\text{-step}})}{\text{median}(\mathcal{R}_K)} \geq 2.0 \right\} \quad (6)$$

For Push-T, this autonomously selects an action-chunked horizon $K^* = 5$ (where displacement $\sim 5\text{--}6 \gg 1.22$).

2. **Topological Identification Cutoff (τ_{id}):** Pairs whose latent separation is smaller than the baseline manifold jitter represent identical or touching physical states (such as the 2.1 px Push-T pairs). Forcing an active transition onto them causes dynamic overshoot. We calibrate:

$$\tau_{\text{id}} = Q_{0.15}(\mathcal{D}_{\text{step}}) \quad (7)$$

(the 15th percentile of 1-step displacement; $\tau_{\text{id}} \approx 0.88$ on Push-T). Any candidate pair with $\|\mathbf{z}_i - \mathbf{z}_j\| \leq \tau_{\text{id}}$ is classified as a **0-cost topological identification edge** ($i \equiv j$) without querying the IDM or predictor.

3. **Dynamic Reachability Search Radius (τ_{reach}):** The upper bound on candidate distance for single-transition queries is set to the 95th percentile of empirical K^* -step transitions:

$$\tau_{\text{reach}} = Q_{0.95}(\mathcal{D}_{K^*\text{-step}}) \quad (8)$$

On Push-T ($K^* = 1$), $\tau_{\text{reach}} \approx 2.40$. Any pair with $\|\mathbf{z}_i - \mathbf{z}_j\| > \tau_{\text{reach}}$ is physically unreachable in a single transition and is pruned before model inference.

4. **IDM Epistemic Uncertainty Gate (τ_{var}):** The ensemble variance threshold separates kinematically viable transitions from out-of-distribution hallucinations:

$$\tau_{\text{var}} = Q_{0.95}(\mathcal{V}_{\text{in}}) \quad (9)$$

Cross-episode candidate transitions with $\sigma_{\mathcal{I}}^2(\mathbf{z}_i, \mathbf{z}_j) > \tau_{\text{var}}$ are rejected as out-of-distribution.

5. **Predictor Relative Consistency Ratio (τ_{rel}):** Instead of thresholding absolute predictor error (which varies wildly across embeddings), we evaluate whether the IDM-proposed action explains state evolution substantially better than idling:

$$\rho(i \rightarrow j) = \frac{\|\mathcal{P}(\mathbf{z}_i, \bar{a}_{i \rightarrow j}) - \mathbf{z}_j\|_2}{\|\mathcal{P}(\mathbf{z}_i, a_{\text{null}}) - \mathbf{z}_j\|_2} \quad (10)$$

The acceptance threshold is calibrated against the 85th percentile of real transitions:

$$\tau_{\text{rel}} = \min(0.90, Q_{0.85}(\mathcal{Q}_{\text{rel}})) \quad (11)$$

A candidate transition $i \rightarrow j$ is accepted if and only if $\rho(i \rightarrow j) \leq \tau_{\text{rel}}$.

6. **Bidirectional Asymmetric Evaluation:** Because physical contact and velocity vectors are directed, candidate pairs (i, j) in the active band $(\tau_{\text{id}}, \tau_{\text{reach}}]$ are tested independently in both directions: $i \rightarrow j$ (using $\mathcal{I}(\mathbf{z}_i, \mathbf{z}_j)$) and $j \rightarrow i$ (using $\mathcal{I}(\mathbf{z}_j, \mathbf{z}_i)$). Graph edges are inserted as directed transitions matching verified physical feasibility.

5.3 Master Algorithm: Self-Calibrating Latent Graph Construction

The complete operational pipeline is formalized in Algorithm 1. The algorithm operates in two phases: an offline calibration phase that runs once on the logged dataset $\mathcal{D}$, and an online graph construction phase that assembles landmarks, builds a k -NN index, and inserts calibrated topological and dynamic edges.

Algorithm 1 Self-Calibrating Latent Graph Construction (Strictly Latent-Only)

Require: Offline visual latent dataset $\mathcal{D} = \{(\mathbf{z}_t^{(e)}, a_t^{(e)})\}$, trained visual predictor $\mathcal{P}$, trained IDM ensemble $\{\mathcal{I}_m\}_{m=1}^M$, landmark count N_{LM} , kNN neighbors k .

Ensure: Calibrated directed topological graph $G = (V, E, W, A)$ with vertices V , directed edges E , weights W , and transition actions A .

// **Phase 1: Autonomous Dataset Calibration (No Proprioception)**

- 1: Sample N_{calib} consecutive transitions $\{(\mathbf{z}_t, a_t, \mathbf{z}_{t+1})\}$ from $\mathcal{D}$.
- 2: Compute displacement distribution $\mathcal{D}_{\text{step}}$ and 1-step predictor residual distribution $\mathcal{R}_1$.
- 3: Compute $\text{SNR}(1) \leftarrow \text{median}(\mathcal{D}_{\text{step}}) / \text{median}(\mathcal{R}_1)$.
- 4: **if** $\text{SNR}(1) \geq 2.0$ **then**
- 5: $K^* \leftarrow 1$ ▷ High SNR regime (e.g., Two-Room)
- 6: **else**
- 7: $K^* \leftarrow \min\{K \in \{2, 4, 5, 8, 16\} \mid \text{SNR}(K) \geq 2.0\}$ ▷ Low SNR regime (e.g., Push-T)
- 8: **end if**
- 9: Compute cutoffs from empirical quantiles:
- 10: $\tau_{\text{id}} \leftarrow Q_{0.15}(\mathcal{D}_{\text{step}})$ ▷ Topological identification threshold
- 11: $\tau_{\text{reach}} \leftarrow Q_{0.95}(\mathcal{D}_{K^* \text{-step}})$ ▷ Maximum candidate search radius
- 12: $\tau_{\text{var}} \leftarrow Q_{0.95}(\mathcal{V}_{\text{in}})$ ▷ IDM ensemble uncertainty gate
- 13: $\tau_{\text{rel}} \leftarrow \min(0.90, Q_{0.85}(\mathcal{Q}_{\text{rel}}))$ ▷ Relative predictor verification gate

// **Phase 2: Landmark Selection and Trajectory Backbone**

- 14: Sample N_{LM} landmark vertices $V = \{\mathbf{z}_1, \dots, \mathbf{z}_{N_{\text{LM}}}\}$ from $\mathcal{D}$ (via FPS or uniform stratum).
- 15: Initialize $E \leftarrow \emptyset$, $W \leftarrow \emptyset$, $A \leftarrow \emptyset$.
- 16: **for** each consecutive pair $(\mathbf{z}_t, \mathbf{z}_{t+1})$ in dataset trajectories present in V **do**
- 17: Add directed edge $(t, t+1)$ with weight $w = 1.0$ and action $a = a_t$. ▷ In-trajectory backbone
- 18: **end for**

// **Phase 3: Cross-Episode Topological and Directed Transition Stitching**

- 19: Build k -NN index on V using Euclidean latent metric $d(\mathbf{z}_i, \mathbf{z}_j)$.
- 20: **for** each landmark $\mathbf{z}_i \in V$ **do**
- 21: **for** each neighbor $\mathbf{z}_j \in \text{kNN}(\mathbf{z}_i, k)$ with $\text{ep}(i) \neq \text{ep}(j)$ **do**
- 22: $d_{ij} \leftarrow \|\mathbf{z}_i - \mathbf{z}_j\|_2$
- 23: **if** $d_{ij} \leq \tau_{\text{id}}$ **then**
- 24: // **Tier 1: Topological Identification Edge (Zero Cost)**
- 25: Add bidirectional edges (i, j) and (j, i) to E with weight $w = 0$ and action $a = \mathbf{0}$.
- 26: **else if** $d_{ij} \leq \tau_{\text{reach}}$ **then**
- 27: // **Tier 2: Dynamic Directed Transition Edge (Unit Cost)**
- 28: **for** direction $(u, v) \in \{(i, j), (j, i)\}$ **do**
- 29: Query IDM ensemble: $\bar{a}_{u \rightarrow v} \leftarrow \frac{1}{M} \sum_{m=1}^M \mathcal{I}_m(\mathbf{z}_u, \mathbf{z}_v)$.
- 30: Compute variance $\sigma_{\mathcal{I}}^2(u, v) \leftarrow \frac{1}{M} \sum_{m=1}^M \|\mathcal{I}_m(\mathbf{z}_u, \mathbf{z}_v) - \bar{a}_{u \rightarrow v}\|_2^2$.
- 31: **if** $\sigma_{\mathcal{I}}^2(u, v) \leq \tau_{\text{var}}$ **then**
- 32: $\text{err}_{\text{idm}} \leftarrow \|\mathcal{P}(\mathbf{z}_u, \bar{a}_{u \rightarrow v}) - \mathbf{z}_v\|_2$
- 33: $\text{err}_{\text{null}} \leftarrow \|\mathcal{P}(\mathbf{z}_u, a_{\text{null}}) - \mathbf{z}_v\|_2$
- 34: **if** $\text{err}_{\text{idm}} / \text{err}_{\text{null}} \leq \tau_{\text{rel}}$ **then**
- 35: Add directed edge (u, v) to E with weight $w = 1.0$, action $a = \bar{a}_{u \rightarrow v}$.
- 36: **end if**
- 37: **end if**
- 38: **end for**
- 39: **end if**
- 40: **end for**
- 41: **end for**
- 42: **return** Calibrated graph $G = (V, E, W, A)$.

5.4 Failure-Mode Defense Matrix

Table 4 summarizes every empirical failure mode uncovered during the Two-Room and Push-T investigations, the underlying physical mechanism, the resulting latent symptom, and the corresponding automated architectural safeguard implemented in Algorithm 1.

Table 4: Failure-Mode Defense Matrix: Mapping observed physical anomalies to autonomous latent safeguards.

Failure Mode	Physical Mechanism	Latent Symptom	Automated Defense	Empirical Proof
Near-Duplicate Overshoot	End-effector already touching or proximate to target (2.1 px). Active step overshoots.	Small latent distance ($d < 1.0$). IDM action increases physical distance on 85.9% of pairs.	Topological Identification Cutoff: Assign weight $w = 0$, action $\mathbf{0}$ for $d \leq \tau_{\text{id}} = Q_{0.15}(\mathcal{D}_{\text{step}})$.	Eliminates overshooting queries entirely (Table 3).
Predictor Noise Floor Compression	Single-step displacement (8.3 px $\rightarrow$ 1.15 latent) is smaller than predictor error (1.22).	Latent IDM-to-random ratio compressed to 1.019 $\times$ despite 80%–89% physical superiority.	SNR Horizon Scaling (K^*): Shift from 1-step to K -step chunked horizon where $\text{SNR}(K) \geq 2.0$.	Restores SNR from 0.94 to > 4.5 on Push-T (Section 4.5).
Directed Irreversibility	Contact dynamics are non-reversible (pushing a block cannot be inverted by reversing pusher velocity).	Symmetric indexing ($i < j$) drops valid transitions oriented from $j \rightarrow i$.	Bidirectional Directed Evaluation: Test $i \rightarrow j$ and $j \rightarrow i$ independently. Insert directed edges.	Increases cross-episode connectivity from 48.3% to 63.4% (Section 4.4).
Out-of-Distribution Hallucination	k -NN selects geometrically close states separated by unbridgeable physical obstacles or configurations.	IDM ensemble heads diverge significantly; epistemic variance spikes.	Epistemic Uncertainty Gate: Prune candidate if $\sigma_{\mathcal{I}}^2 > \tau_{\text{var}} = Q_{0.95}(\mathcal{V}_{\text{in}})$.	Rejects non-viable extrapolation pairs while preserving 95% of valid transitions.
Dynamic Inefficacy (Null Drift)	Proposed action does not move state toward target any faster than idle drift or system inertia.	Predictor error under $\bar{a}_{\text{idm}}$ equals or exceeds error under a_{null} .	Relative Predictor Ratio: Enforce $\rho(u \rightarrow v) \leq \tau_{\text{rel}} = \min(0.90, Q_{0.85}(\mathcal{Q}_{\text{rel}}))$.	IDM beats random on 79.9%–88.6% of candidates in step and distant regimes (Table 3).